

HouseHunt - Real Estate Application

BraintStorming & Idea Prioritisation

HouseRentals

Phase : BraintStorming & Idea Prioritisation

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GitHub Repository Link: [Click Here](#)

Date	18-06-2026
Project Name	HouseRentals
Maximum Marks	4Marks

1. PROJECT OVERVIEW

HouseHunt is a comprehensive digital marketplace designed to revolutionize real estate transactions. The platform connects property seekers, owners, and real estate professionals through an intuitive interface powered by advanced search capabilities, real-time market insights, and seamless communication tools.

Core Objectives:

- Create a user-centric platform for property discovery and transactions
- Enable seamless communication between renters/buyers and landlords/sellers
- Provide market insights and property analytics
- Support real estate professionals with client management tools
- Ensure 24/7 accessibility across web and mobile platforms

2. KEY FEATURES & FUNCTIONALITIES

Feature	Description
Advanced Search	Filter properties by location, price, amenities, property type
Virtual Tours	360° property tours and high-resolution image galleries
Property Listings	Detailed property information with owner/agent contact
User Profiles	Buyer/Seller/Agent profiles with ratings and reviews
Wishlist/Favorites	Save properties for later viewing
Market Analytics	Price trends, neighborhood insights, market reports
Document Management	Upload and manage legal documents

3. USER PERSONAS

Property Seeker/Renter: Looking for rental or purchase properties with specific requirements

Property Owner/Landlord: Wanting to list and manage rental properties

Real Estate Agent: Managing client listings and facilitating transactions

Investor: Analyzing property data and market trends for investment decisions

4. MERN STACK ARCHITECTURE

MongoDB - Database Layer

NoSQL document database for flexible data storage. Collections for users, properties, listings, transactions, messages, and reviews.

Express.js - Backend Framework

RESTful API server handling authentication, property management, search, messaging, and payment integration.

React - Frontend Framework

Component-based UI with Redux for state management, responsive design for desktop and mobile.

Node.js - Runtime Environment

JavaScript runtime for server-side execution, enabling full-stack JavaScript development.

5. DATABASE SCHEMA DESIGN (MongoDB)

Core Collections:

Users: id, name, email, password, phone, role (buyer/seller/agent), profile_pic, rating, reviews

Properties: id, title, description, address, price, amenities, photos[], location (coords), owner_id

Bookings : bookingId, UserID, PropertyOwnerId, amount, userName, availableStatus.

6. REACT FRONTEND COMPONENTS

Authentication Pages:

- Login
- Register
- Password Reset
- Profile Setup and More

Main Pages:

- Home/Dashboard
- Property Search
- Property Details
- User Profile

Common Components:

- Navigation Bar
- Image Gallery

7. EXPRESS.JS API ENDPOINTS

Authentication: POST /api/auth/register, POST /api/auth/login, POST /api/auth/logout

Users: GET /api/users/:id, PUT /api/users/:id, GET /api/users/:id/listings

Properties: GET /api/properties, POST /api/properties, GET /api/properties/:id, PUT /api/properties/:id

Search: GET /api/search?filters

Bookings: GET /api/ownerBookings/:id, POST /api/CandleBooking.

Admin: GET /api/Allusers GET /api/allBookings DELETE /api/deleteUser

8. KEY TECHNOLOGIES & DEPENDENCIES

Backend:

- Express.js
- Mongoose (MongoDB ODM)
- JWT (Authentication)

- Bcrypt (Password Hashing)
- Multer (File Upload)
- Nodemailer

Frontend:

- React
- Axios
- TailwindCS

9. KEY USER FLOWS

Property Search Flow:

- 1. User enters search filters (location, price, amenities)
- 2. Backend queries MongoDB based on filters
- 3. Results displayed with property cards
- 4. User clicks property for detailed view
- 5. User can message owner/agent or add to favorites

Property Listing Flow:

- 1. Owner/Agent logs in and navigates to 'Create Listing'
- 2. Fills property details (title, description, amenities, price)
- 3. Uploads property images and documents
- 4. Sets property location on map
- 5. Publishes listing to marketplace
- 6. Tracks views and interactions

Messaging Flow:

- 1. Interested user initiates conversation
- 2. Real-time message delivery via Socket.io
- 3. Both parties can exchange details and negotiate
- 4. Messages persist in database

10. IMPLEMENTATION TIMELINE

Phase	Duration	Tasks
Phase 1: Setup	Week 1 -2	Project setup, DB schema, API structure
Phase 2: Backend	Weeks 1-2	Auth, User, Property APIs
Phase 3: Frontend	Weeks 3	UI components, pages, integration
Phase 4: Testing	Week 4	Bug fixes, optimization

11. POTENTIAL CHALLENGES & SOLUTIONS

Challenge: Image Management

Solution: Implement cloud storage (AWS S3, Cloudinary) with CDN

Challenge: Search Performance

Solution: Create MongoDB indexes on frequently searched fields

Challenge: Data Validation

Solution: Implement validation on both frontend and backend

12. FUTURE ENHANCEMENTS (PHASE 2)

- Mobile app (React Native)
- AI-powered property recommendations
- Virtual property tours with VR support
- Advanced analytics dashboard for agents
- Integration with local property databases
- Video calling between users
- Automated document generation
- Loan calculator and mortgage tools
- Insurance integration
- Multi-language support

13. SUCCESS METRICS

- User registration and active users
- Number of property listings
- Average response time for API calls
- User satisfaction rating
- Page load time (<3 seconds)
- Mobile responsiveness

